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Collaborative truck multi-drone routing and scheduling problem: Package delivery with flexible launch and recovery sites

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ABSTRACT

This paper deals with the problem of coordinating a truck and multiple heterogeneous unmanned aerial vehicles (UAVs or drones) for last-mile package deliveries. Existing literature on truck–drone tandems predominantly restricts the UAV launch and recovery operations (LARO) to customer locations. Such a constrained setting may not be able to fully exploit the capability of drones. Moreover, this assumption may not accurately reflect the actual delivery operations. In this research, we address these gaps and introduce a new variant of truck–drone tandem that allows the truck to stop at non-customer locations (referred to as flexible sites) for drone LARO. The proposed variant also accounts for three key decisions — (i) assignment of each customer location to a vehicle, (ii) routing of truck and UAVs, and (iii) scheduling drone LARO and truck operator activities at each stop, which are always not simultaneously considered in the literature. A mixed integer linear programming model is formulated to jointly optimize the three decisions with the objective of minimizing the delivery completion time (or makespan). To handle large problem instances, we develop an optimization-enabled two-phase search algorithm by hybridizing simulated annealing and variable neighborhood search. Numerical analysis demonstrates substantial improvement in delivery efficiency of using flexible sites for LARO as opposed to the existing approach of restricting truck stop locations. Finally, several insights on drone utilization and flexible site selection are provided based on our findings.

1. Introduction

The last-mile delivery market is projected to have a compounded annual growth rate of 16% during the next five years, primarily due to the proliferation of e-commerce transactions (Reuters, 2021). Consequently, the traffic congestion and greenhouse gas emissions are expected to increase by 21% and 32%, respectively, in the absence of an effective last-mile intervention strategy (World Economic Forum, 2020). On the other hand, companies are also striving to meet the faster delivery expectations of the consumer and gain a competitive advantage. A potential approach to transform the last-mile delivery is to adopt emerging logistics technologies, namely unmanned aerial vehicles (UAVs) or drones, as they can circumvent ground-based traffic congestion and parallelize delivery operations (Lemardelé et al., 2021). In particular, a drone can surmount its limited payload capacity and flying range by collaborating with a truck that can serve as a moving depot and recharging (or battery-swap) platform (Hong et al., 2018; Karak and Abdelghany, 2019; Jeong et al., 2019; Moshref-Javadi et al., 2020; Salama and Srinivas, 2020; Bruni et al., 2022; Leon-Blanco et al., 2022).

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While recent literature has extensively focused on truck–drone routing, most of them restrict the drone launch and recovery operations (LARO) to customer locations (Yurek and Ozmutlu, 2018; Agatz et al., 2018; Wang and Sheu, 2019; Murray and Raj, 2020; Moshref-Javadi et al., 2020). However, the truck can stop at other feasible locations in the service region to deploy or recover the drones. As a matter of fact, real-life testing by UPS shows the drone being launched from the roof of a delivery truck at non-customer locations (UPS, 2017). Moreover, scheduling of truck operator tasks at a given stop, which possibly includes customer service and launch/recovery of multiple drones, is usually neglected (Chang and Lee, 2018; Schermer et al., 2019a; Poikonen and Golden, 2020). Nevertheless, it is crucial to consider the sequencing dynamics to ensure safe and realistic operations (Poikonen and Campbell, 2021).

An illustrative example is provided in Fig. 1 to highlight both the potential of allowing non-customer nodes (also referred to as flexible locations or sites) for drone LARO and the importance of scheduling at truck stops. The network in this example consists of a depot, 8 customer locations to be served by a fleet of truck and four drones, and 8 non-customer locations. Note that the flexible LARO sites can be established in advance by considering the feasible areas within the delivery region. It can be observed from the optimal plan that the delivery completion time is substantially lower when flexible stop locations are permitted. In this particular example, the distance traveled by truck is shorter and drone fleet utilization is higher if trucks are allowed to launch and recover drones at flexible locations. For instance, in the case of restricted LARO, node 4 is served by a truck instead of a drone since existing locations along the truck route do not meet the flight range restriction. As a result, the truck route is significantly longer when compared to the case of using flexible sites. Furthermore, while the truck carries four drones in both settings, only three of them were utilized when the LARO are restricted. Hence, the drones play a secondary role in aiding the truck when restricting LARO to customer locations. In contrast, when flexible stop locations are considered, the truck assists the drones for faster delivery completion.

Motivated by the potential and feasibility of flexible locations in practice, this paper aims to minimize the completion time (or makespan) of last-mile delivery operations by employing a mixed fleet (truck and multiple drones) and accounting for flexible truck stops for drone LARO. Specifically, this paper contributes to the research stream of collaborative truck–drone system for package delivery in the following ways.

- A new variant of drone-enabled logistics for last-mile package delivery called the collaborative truck–drone routing and scheduling problem with flexible launch and recovery locations (CTDRSP-FL) is introduced.
- To the best of our knowledge, this is the first work to consider flexible truck stops with the following three logistical decisions: (i) *assignment*: whether a drone or truck serves a specific location, (ii) *routing*: in which order should the truck visit the assigned locations (truck routing), and at which truck stop should a drone be launched and recovered to serve a customer (drone routing or sortie), and (iii) *scheduling*: launch and retrieval times of multiple drones at a given truck stop. Unlike the prior works, the truck–drone fleet is not required to visit all the nodes because certain flexible stop locations may never be visited (see unselected flexible locations in Fig. 1). Besides, incorporating the scheduling of drones at each truck stop further complicates the problem.
- A fleet of heterogeneous drones characterized by their endurance and speed is considered. As drone technology is expected to evolve rapidly, companies are bound to have UAVs of various capabilities (e.g., quadcopter vs. fixed-wing drones), and it becomes critical to account for these attributes during route planning.
- A new mixed integer linear programming (MILP) model is developed to handle the flexible locations for drone LARO and simultaneously optimize the three aforementioned logistical decisions.
- An optimization-enabled two-phase search (OTS) algorithm is developed to solve large size instances in a reasonable time. Specifically, simulated annealing (SA) algorithm is employed in the first phase to explore specific areas of the solution space, and a variable neighborhood search (VNS) algorithm expands the search in the second phase by leveraging the solution obtained by the SA algorithm. Besides, a modular approach is adopted as it provides greater flexibility to handle evolving regulations and drone characteristics, while also enabling scalable deployment.
- Several new test instances are introduced and extensive analysis is conducted to evaluate the proposed models. The delivery completion time associated with the proposed CTDRSP-FL model is benchmarked against the conventional approach of restricting the LARO to be at customer locations.

The remainder of this paper is organized as follows. Section 2 reviews related studies in hybrid truck–drone systems and highlights the gaps in the existing literature. In Section 3, the proposed CTDRSP-FL variant is described and its mathematical formulation is presented. Next, the proposed OTS algorithm for solving large size instances is developed in Section 4. Section 5 presents the numerical analysis that focuses on demonstrating the impact of using flexible locations for LARO and the effectiveness of the proposed solution approach. Finally, the conclusion and directions for future research are provided in Section 6.

2. Literature review

There is a growing body of literature on combined truck–drone operations for last-mile delivery. Interested readers are referred to recent surveys conducted by Khoufi et al. (2019), Chung et al. (2020), Macrina et al. (2020), Benarbia and Kyamakya (2021), and Rojas Viloria et al. (2021) for routing problems with drones. Also, a review from the operational research perspective of drone-aided parcel delivery is provided by Boysen et al. (2021). In this section, we review recent practical developments on drone deliveries, and then focus on research works that employ truck–drone tandems for last-mile deliveries. Furthermore, to highlight the novelty of our study, we also review prior studies that do not restrict drone LARO to customer locations.

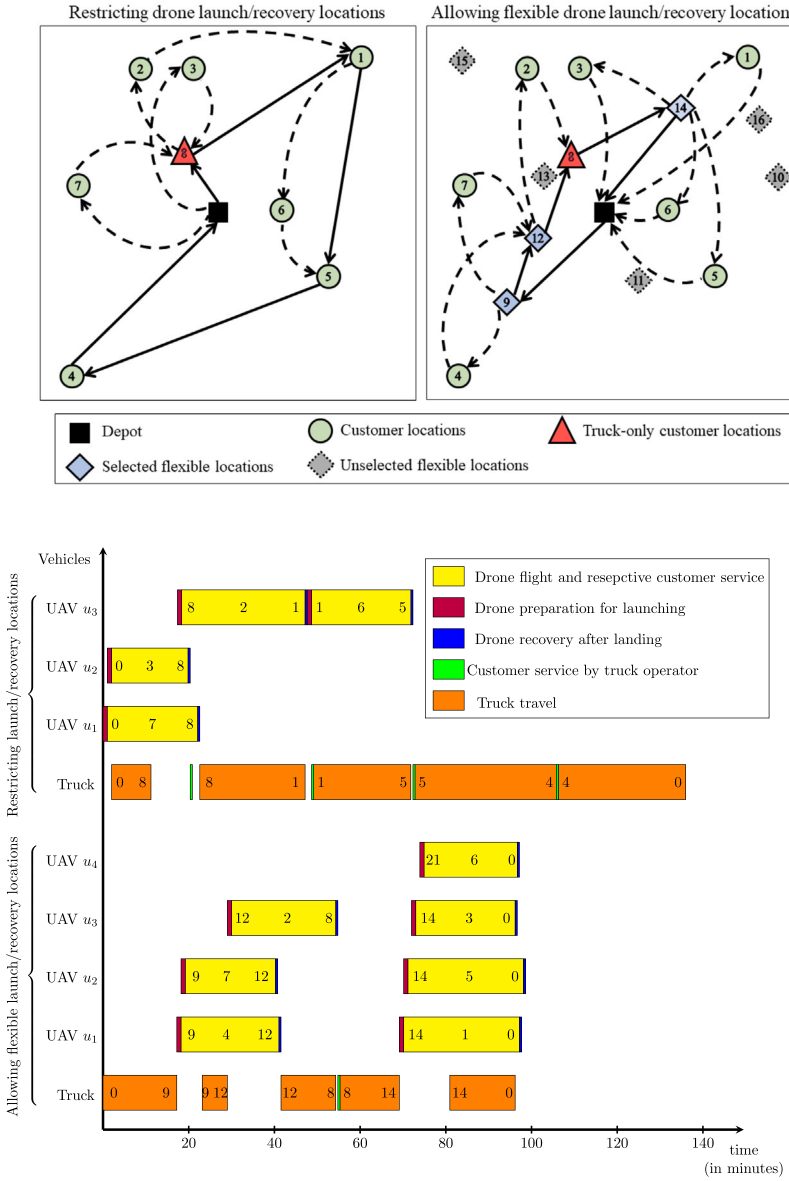

Fig. 1. Illustrative example of using flexible locations for drone launching and recovery. The solid arrows are for truck route and the dashed arcs are for drone sorties. The numbers in the Gantt chart are for the locations on the truck/drone routes.

From a practical perspective, employing drones for last-mile deliveries is extensively tested by many corporations, such as Amazon, Google, Fedex, DHL, and UPS, to be widely deployed in the near future (Murray and Raj, 2020; Chung et al., 2020; Poikonen and Golden, 2020). Furthermore, realizing the potential of hybrid truck–drone systems spurred several companies to develop proof-of-concept technologies for embracing the idea (e.g., Mercedes Vision Van with automated cargo space management and drone deployment (Mercedes-Benz, 2018)). Also, pilot experiments of truck–drone deliveries have been conducted (UPS, 2017; Workhorse, 2020). For example, Workhorse, a delivery company in Ohio, tested residential deliveries using a hybrid truck–drone system (Workhorse, 2020). On the regulatory aspect, several countries across the globe are relaxing rules to adopt commercial drones in the low altitude airspace. For instance, the Federal Aviation Administration (FAA) in the United States is now allowing small-sized drones to fly over people and operate beyond visual line of sight (BVLOS), even at night, thereby removing significant barriers to their widespread use (FAA, 2020). Similarly, the European Union Aviation Safety Agency (EASA) has standardized drone regulations for all its member states, and provides authorization for BVLOS operations (EASA, 2020).

Owing to the complexity of hybrid truck–drone systems, a lot of academic research in recent years has focused on optimizing the collaborative route plans, especially by extending classical routing problems, such as the traveling salesman problem (TSP) and vehicle routing problem (VRP) (Murray and Chu, 2015; Bouman et al., 2018; Agatz et al., 2018; Sacramento et al., 2019; Wang and

Table 1
Summary of recent relevant works on hybrid truck–drone delivery systems.

References	UAV Characteristics		LARO of UAVs[a]					Decisions pertaining to truck–drone deliveries		
	Multiple UAVs	Heterogeneous UAVs	C	N	E	A	F	Assignment	Routing	Scheduling
Murray and Chu (2015)	–	N/A	✓	–	–	–	–	✓	✓	N/A
Agatz et al. (2018)	–	N/A	✓	–	–	–	–	✓	✓	N/A
Yurek and Ozmutlu (2018)	–	N/A	✓	–	–	–	–	✓	✓	N/A
Bouman et al. (2018)	–	N/A	✓	–	–	–	–	✓	✓	N/A
Dell'Amico et al. (2019)	–	N/A	✓	–	–	–	–	✓	✓	N/A
Murray and Raj (2020)	✓	✓	✓	–	–	–	–	✓	✓	✓
Cavani et al. (2021)	✓	–	✓	–	–	–	–	✓	✓	–
Luo et al. (2021)	✓	–	✓	–	–	–	–	✓	✓	✓
Sacramento et al. (2019)	✓	–	✓	–	–	–	–	✓	✓	N/A
Schermer et al. (2019b)	✓	–	✓	–	–	–	–	✓	✓	–
Tamke and Buscher (2021)	✓	–	✓	–	–	–	–	✓	✓	✓
Wang and Sheu (2019)	✓	–	–	✓	–	–	–	✓	✓	N/A
Poikonen and Golden (2020)	✓	–	–	✓	–	–	–	–	✓	–
Ferrandez et al. (2016)	✓	–	–	✓	–	–	–	–	✓	–
Chang and Lee (2018)	✓	–	–	✓	–	–	–	–	✓	–
Karak and Abdelghany (2019)	✓	–	–	✓	–	–	–	–	✓	–
Gonzalez-R et al. (2020)	–	N/A	–	–	✓	–	–	✓	✓	N/A
Marinelli et al. (2018)	–	N/A	–	–	✓	–	–	✓	✓	N/A
Carlsson and Song (2018)	–	N/A	–	–	✓	–	–	–	✓	N/A
Schermer et al. (2019a)	✓	–	–	–	✓	–	–	✓	✓	–
Salama and Srinivas (2020)	✓	–	–	–	–	✓	–	✓	✓	–
This paper	✓	✓	–	–	–	–	✓	✓	✓	✓

[a] C: restricted to customer locations; N: restricted to non-customer locations; E: restricted to en-route of truck; A: anywhere in the delivery region (possibly including infeasible locations); F: flexible locations (i.e., both customer and feasible non-customer locations).

Sheu, 2019; Jackson and Srinivas, 2021). Specifically, the traveling salesman problem with drone (TSP-D) considers a single truck visiting a subset of customer locations, while the remaining customers are served by a drone (Murray and Chu, 2015; Mathew et al., 2015; Agatz et al., 2018; Bouman et al., 2018; Yurek and Ozmutlu, 2018; Ha et al., 2018; Jeong et al., 2019; Gonzalez-R et al., 2020). The drone takes off from a stop, visits a customer location, and then rendezvous with the truck at the same or different location. Nevertheless, the truck stops are always restricted to customer locations. In contrast, the vehicle routing problem with drones (VRP-D) considers coordination between multiple trucks carrying drones to complete the delivery operations (Wang et al., 2017; Poikonen et al., 2017; Ham, 2018; Sacramento et al., 2019; Schermer et al., 2019b; Tamke and Buscher, 2021; Wang and Sheu, 2019). While the aforementioned studies in both TSP-D and VRP-D seek to effectively optimize the last-mile delivery operations under different circumstances and assumptions, all of them have commonly assumed the drone LARO to be restricted at customer locations or stationary docking stations. On the other hand, few existing works on truck–drone routing allow the truck stops (and LARO of drones) to be at non-customer locations, but without integrating the aforementioned three logistical decisions (Ferrandez et al., 2016; Chang and Lee, 2018; Marinelli et al., 2018; Carlsson and Song, 2018; Schermer et al., 2019a; Salama and Srinivas, 2020; Poikonen and Golden, 2020; Gomez-Lagos et al., 2021). A comparison between our work and relevant recent papers is conducted in Table 1 in terms of characteristics associated with UAVs, LARO, and logistical decisions.

As shown in Table 1, the seminal research works in truck–drone systems studied employing a single drone in tandem with truck (Murray and Chu, 2015; Agatz et al., 2018; Yurek and Ozmutlu, 2018; Bouman et al., 2018; Dell'Amico et al., 2019), while subsequent researchers gave more attention to using a fleet of multiple drones (Murray and Raj, 2020; Wang and Sheu, 2019; Schermer et al., 2019b; Sacramento et al., 2019; Poikonen and Golden, 2020; Luo et al., 2021; Cavani et al., 2021; Tamke and Buscher, 2021). Nevertheless, a homogeneous fleet of drones was predominantly assumed, despite the practical need of using heterogeneous drones as operators always periodically invest in their assets (Murray and Raj, 2020). Furthermore, Table 1 exhibits five categories of LARO sites, highlighting the growing research interest in expanding the problem network beyond the depot and customer locations. It is evident that the early research studies on hybrid truck–drone systems were influenced by classical routing problems, such as TSP and VRP, in restricting LARO sites to customer locations (Murray and Chu, 2015; Agatz et al., 2018; Yurek and Ozmutlu, 2018; Bouman et al., 2018; Dell'Amico et al., 2019; Murray and Raj, 2020; Sacramento et al., 2019; Luo et al., 2021; Cavani et al., 2021; Tamke and Buscher, 2021). In contrast, some other studies constrained LARO sites to non-customer locations (Wang and Sheu, 2019; Poikonen and Golden, 2020; Ferrandez et al., 2016; Chang and Lee, 2018; Karak and Abdelghany, 2019). For instance, Wang and Sheu (2019) formulated an arc-based model for the VRP-D with multiple drones, but assumed that the drones can land only at service hubs or the depot, not customer locations, and the supply of drones at such hubs is unlimited. Ferrandez et al. (2016) partitioned the customer locations into clusters using the k-means algorithm — a well-known unsupervised machine learning approach. Then, the launching and recovery of drones are undertaken at the center of each cluster, which is typically a non-customer location as all customers are assumed to be served by drones. A similar approach is used by Chang and Lee (2018) while the locations of truck stops, represented by the focal points of clusters, are optimized to minimize the system-wide delivery completion time.

Another innovative approach for improving hybrid truck–drone operations is to allow drone LARO to be along the truck route (referred to as en-route operations) (Gonzalez-R et al., 2020; Marinelli et al., 2018; Carlsson and Song, 2018; Schermer et al., 2019a; Thomas et al.). Marinelli et al. (2018) developed a greedy heuristic altering a TSP-D routing solution by later launching (relative to the originally planned) and early retrieval of drones on the truck route. Carlsson and Song (2018) considered drones that can land on the truck roof while it is moving. They performed a combination of continuous approximation analysis and numerical simulations to demonstrate the efficiency of en-route LARO in improving delivery operations. Similarly, Schermer et al. (2019a) developed a hybrid variable neighborhood search and tabu search algorithm for the VRP-D with identical drones. For each arc connecting a pair of customers with possible truck travel, they generated potential drone launching and recovery locations. However, in practice, the trucks may be prohibited from stopping at some locations on their routes or cannot park safely (e.g., on a highway). Therefore, restricting those potential locations to be along the truck routes may not achieve the best possible routing plan. The locations for LARO were further relaxed by Salama and Srinivas (2020) by allowing them to be anywhere in the delivery region (including customer and non-customer locations). However, they assumed that drones are recovered at the same launching location after completing their delivery tasks. Moreover, it may not be feasible for a truck to stop at any point in the delivery region, such as private properties or natural areas (e.g., rivers, forests, and mountains). Using flexible sites for UAV operations has the potential to overcome such shortcomings as the delivery network can be expanded to include only feasible non-customer locations for LARO.

Most prior works optimize only the assignment and routing decisions pertaining to the combined truck–drone operations, as shown in Table 1. Moreover, a few studies even dismissed the assignment decision by assuming serving all customers by drones (Ferrandez et al., 2016; Carlsson and Song, 2018; Chang and Lee, 2018; Karak and Abdelghany, 2019; Poikonen and Golden, 2020), while the majority considered that the truck can directly participate in package delivery, especially due to technical reasons such as limited payload capacity of drones (Salama and Srinivas, 2020). However, the scheduling decision is rarely considered in the existing literature (Poikonen and Campbell, 2021). For example, the interaction between sorties of drones was neglected in prior works by computing the delivery time as the summation of maximum travel time of vehicles on subsequent truck travel segments (Schermer et al., 2019b; Poikonen and Golden, 2020; Cavani et al., 2021). Nevertheless, it is crucial to consider the scheduling dynamics to ensure safe and realistic delivery operations. The significance of such interaction can be observed from the Gantt chart in Fig. 1. This research addresses the aforementioned gaps and is the first to consider flexible sites for drone LARO while optimizing the scheduling of drone launching and landing, as well as customer service, at truck stops. In contrast to the en-route LARO, the flexible sites are not assumed to be located along the truck path between two customer nodes. Thus, a holistic approach is adopted to model and solve the problem under consideration. Our goal in this paper is to demonstrate the potential impact of using flexible locations as truck stops for drone launching and recovery, thereby urging future research works to consider such locations for improving their last-mile delivery routing solutions.

3. Problem description and mathematical formulation

This research considers the last-mile delivery of packages to a set of customer locations $\mathcal{N} = \{i_1, i_2, i_3, \dots i_N\}$ distributed in a 2D Euclidean space using a truck and a set of heterogeneous drones $\mathcal{U} = \{u_1, u_2, u_3, \dots u_U\}$. While all locations are typically reachable by truck, drones can only serve a subset of locations $\mathcal{N}^U \subseteq \mathcal{N}$ due to technical and operational restrictions (e.g., limited payload capacity), and therefore the remaining customers $\mathcal{N}^T \subseteq \mathcal{N}$ (where $\mathcal{N}^T = \mathcal{N} \setminus \mathcal{N}^U$) must be served by truck. In contrast to most of the last-mile delivery literature, our problem expands the delivery network to include an additional set of F feasible truck stop locations (non-customer locations and reachable by truck) besides the depot and customer locations, thereby providing flexibility for drone LARO. The resultant set of feasible locations can therefore be denoted by $\mathcal{F} = \{i_0\} \cup \mathcal{N} \cup \{i_{N+1}, i_{N+2}, i_{N+3}, \dots i_{N+F}\} \cup \{i_{N+F+1}\}$, where i_0 and i_{N+F+1} represent the starting and ending depot, respectively, which can be distinct or same location. Among the $N+F$ feasible locations, up to $\hat{S}$ of customer and non-customer locations can be visited by the truck. Therefore, a set of potential (or selected) truck stops $\mathcal{S} = \{s_0, s_1, s_2, s_3, \dots s_{\hat{S}}, s_{\hat{S}+1}\}$ is considered, which includes the $\hat{S}$ locations in addition to two stops s_0 and $s_{\hat{S}+1}$ corresponding to the depot indices i_0 and i_{N+F+1} in set $\mathcal{F}$, respectively.

The truck carrying packages and a fleet of drones must start from the depot, visit a subset of feasible locations, and return to the depot. The truck travel time from location i to $j \in \mathcal{F}$ is denoted by R^T_{ij}. All customer locations that coincide with the truck route are assumed to be served by the truck operator, requiring service time V^T_i for customer $i \in \mathcal{N}$. Besides, the truck serves as a mobile drone platform for battery swap (or recharging) and LARO. Each UAV $u \in \mathcal{U}$ is allowed to carry one package per trip, where each UAV sortie consists of launching from a truck stop $i \in \mathcal{F}$, visiting a customer location $j \in \mathcal{N}^U$ to deliver the package, and returning to the next truck stop $k \in \mathcal{F}$ requiring flying time R^U_{ijku} and service time V^U_j. All customers must be served either by a drone $u \in \mathcal{U}$ or truck before the fleet returns to the depot. Without loss of generality, the total time of a drone $u \in \mathcal{U}$ between its launching and landing must be within its endurance E_{ijku} for the task from location i to customer j then to location $k \in \mathcal{F}$. Before a drone sortie, the truck operator spends a duration H_i at truck stop (or depot) $i \in \mathcal{F}$ to retrieve the respective order and prepare the drone for launching. Likewise, after completing an order delivery by drone, it requires a recovery time G_i when landing at truck stop $i \in \mathcal{F}$. Thus, overall delivery makespan is the total time between starting the first vehicle preparation to depart from the depot and completing the last vehicle recovery at the depot. Given the aforementioned problem characteristics, the objective of this research is to minimize the delivery completion time.

The problem addressed in this paper is termed as the collaborative truck–drone routing and scheduling problem with flexible launch and recovery locations (CTDRSP-FL). There are three categories of decisions to be made for solving the CTDRSP-FL. First, a vehicle (truck or drone) should be assigned to visit/serve each customer. In addition, truck stops, either customer location or flexible site, must be selected, and then truck route must be established. Furthermore, launching locations of drones (which must be the depot or truck stops) to serve their assigned customers are determined, while the landing of drones are assumed to be at the subsequent truck stop (Carlsson and Song, 2018; Poikonen and Golden, 2020). Second, the departure and arrival times of all vehicles at each truck stop and depot must be determined. Third, since the truck operator is assumed to perform only one task at a time, sequencing (or scheduling) of tasks at truck stops (i.e., drone launching and landing, and customer service) should be ascertained. An MILP is developed to jointly optimize the three decisions and solve the CTDRSP-FL. The following notation are used to formulate the optimization model.

Indices and sets

$i, j, k \in \mathcal{N}$	set of customer locations, $\mathcal{N} = \{i_1, i_2, i_3, \dots i_N\}$
$i, j, k \in \mathcal{N}^U$	set of customer locations that can be served by drone or truck, $\mathcal{N}^U \subseteq \mathcal{N}$
$i, j, k \in \mathcal{N}^T$	set of customer locations that must be served by truck, $\mathcal{N}^T = \mathcal{N} \backslash \mathcal{N}^U$
$i, j, k \in \mathcal{F}$	set of potential truck stops, $\mathcal{F} = \{i_0\} \cup \mathcal{N} \cup \{i_{N+1}, i_{N+2}, i_{N+3}, \dots i_{N+F}\} \cup \{i_{N+F+1}\}$
$s \in \mathcal{S}$	set of selected truck stops, $\mathcal{S} = \{s_0, s_1, s_2, s_3, \dots s_{\hat{S}}, s_{\hat{S}+1}\}$
$u \in \mathcal{U}$	set of UAVs (or drones), $\mathcal{U} = \{u_1, u_2, u_3, \dots u_U\}$

Parameters

N	number of customer locations
F	number of feasible truck stops at non-customer locations
$\hat{S}$	maximum number of stops allowed for the truck
U	number of drones
R^T_{ij}	travel time of truck from feasible truck stop i to $j \in \mathcal{F}$
R^U_{ijku}	travel time of drone $u \in \mathcal{U}$ from feasible truck stop $i \in \mathcal{F}$ to visit delivery location $j \in \mathcal{N}^U$ and then returning to feasible truck stop $k \in \mathcal{F}$
V^T_i	service time of truck at delivery location $i \in \mathcal{N}$
V^U_i	service time of drone at delivery location $i \in \mathcal{N}^U$
E_{ijku}	endurance (in time unit) of drone $u \in \mathcal{U}$ to fly from location $i \in \mathcal{F}$ for delivering a package at location $j \in \mathcal{N}^U$ and flying back to location $k \in \mathcal{F}$
H_i	launch preparation time of drone at truck stop $i \in \mathcal{F}$
G_i	recovery time of drone at truck stop $i \in \mathcal{F}$

Decision variables

d^T_s	departure time of the truck from stop $s \in \mathcal{S}$ heading to $s+1 \in \mathcal{S}$
a^T_s	arrival time of the truck to stop $s \in \mathcal{S}$ coming from $s-1 \in \mathcal{S}$
d^U_i	departure (or launching) time of a drone to visit location $i \in \mathcal{N}$
a^U_i	arrival (or landing) time of a drone after visiting location $i \in \mathcal{N}$
b^U_i	service begin time at customer location $i \in \mathcal{N}$ if it is selected to be a truck stop
e^U_i	service end time at customer location $i \in \mathcal{N}$ if it is selected to be a truck stop
w_{ijs}	1 if customer location $i \in \mathcal{N}$ is served (by launching or recovering the respective drone, or by delivery via truck operator) before another location $j \in \mathcal{N}$ at truck stop $s \in \mathcal{S}$, 0 otherwise
x_{ius}	1 if a delivery location $i \in \mathcal{N}$ is served by UAV $u \in \mathcal{U}$ dispatched from truck stop $s \in \mathcal{S}$, 0 otherwise
y_{is}	1 if location $i \in \mathcal{F}$ is assigned as a truck stop $s \in \mathcal{S}$, 0 otherwise
$\hat{t}$	delivery completion time of all vehicles

3.1. Objective function and decision variables

The objective function (1) minimizes the completion time of delivery operations, which is determined by constraints (2) and (3) as the maximum of truck arrival to the depot and the latest recovery of drone after landing (i.e., $\hat{t} = \max_{\forall i \in \mathcal{N}}(a^T_s, a^U_i + \sum_{u \in \mathcal{U}} G_j x_{ius})$, $j = \{j_{N+F+1}\}, s = \{s_{\hat{S}}, s_{\hat{S}+1}\}$). The binary and continuous variables are specified by constraints (4) and (5), respectively.

$$Minimize \quad \hat{t} \tag{1}$$

$S.t.$

$$\hat{t} \geq a^T_s \qquad \forall s \in \{s_{\hat{S}+1}\} \subseteq \mathcal{S} \tag{2}$$

$$\hat{t} \geq a^U_i + \sum_{u \in \mathcal{U}} G_j x_{ius} \qquad \forall i \in \mathcal{N}, j \in \{j_{N+F+1}\} \subseteq \mathcal{N}, s \in \{s_{\hat{S}}\} \subseteq \mathcal{S} \tag{3}$$

$$w_{ijs}, x_{ius}, y_{is} \in \{0, 1\} \qquad \forall i, j \in \mathcal{F}, u \in \mathcal{U}, s \in \mathcal{S} \tag{4}$$

$$d_s^T, a_s^T, d_i^U, a_i^U, b_i^U, e_i^U, \hat{\tau} \in \mathbb{R}_+ \qquad \forall i \in \mathcal{N}, s \in \mathcal{S} \tag{5}$$

3.2. Assignments and routing of hybrid vehicle fleet

Constraints (6)–(13) determine the assignments of truck stops (that can be customer locations or flexible drone launch/recovery locations), assignments of drones to customers (who were not selected to be visited by truck), and the routing of vehicles (both drones and truck).

$$y_{is} = 1 \qquad \forall i \in \{i_0\} \subseteq \mathcal{F}, s \in \{s_0\} \subseteq \mathcal{S} \tag{6}$$

$$y_{is} = 1 \qquad \forall i \in \{i_{N+F+1}\} \subseteq \mathcal{F}, s \in \{s_{\hat{S}+1}\} \subseteq \mathcal{S} \tag{7}$$

$$\sum_{i \in \mathcal{F}} y_{is} = 1 \qquad \forall s \in \mathcal{S} \tag{8}$$

$$\sum_{s \in \mathcal{S}} y_{is} \leq 1 \qquad \forall i \in \mathcal{F} \backslash \{i_{N+F+1}\} \tag{9}$$

$$y_{is} \leq y_{i,s+1} \qquad \forall i \in \{i_{N+F+1}\} \subseteq \mathcal{F}, s \in \mathcal{S} \backslash \{s_{\hat{S}+1}\} \tag{10}$$

$$\sum_{s \in \mathcal{S}} y_{is} = 1 \qquad \forall i \in \mathcal{N}^T \tag{11}$$

$$\sum_{u \in \mathcal{U}} \sum_{s \in \mathcal{S} \backslash \{s_{\hat{S}+1}\}} x_{ius} + \sum_{s \in \mathcal{S}} y_{is} = 1 \qquad \forall i \in \mathcal{N}^U \tag{12}$$

$$\sum_{i \in \mathcal{N}} x_{ius} \leq 1 \qquad \forall u \in \mathcal{U}, s \in \mathcal{S} \tag{13}$$

The starting and ending depots are represented using two different indices (or stops), $\{s_0\}$ and $\{s_{\hat{S}+1}\}$, to account for their spatial distribution. Such a modeling approach provides the flexibility to represent the depot(s) as a single physical location or two distinct nodes in the $2D$ Euclidean delivery region. Constraints (6) and (7) ensure that the truck starts and ends its route at the depot. Constraint (8) guarantees that each selected truck stop $s \in \mathcal{S}$ is assigned to exactly one potential location $i \in \mathcal{F}$, while constraint (9) ensures that each potential stop can be visited at most once by truck. Furthermore, constraint (10) limits the assignments to the depot index $\{j_{N+F+1}\}$ to a consecutive subset of truck stop indices ending with $\hat{S}+1$. Each customer location that must be visited only by truck, $i \in \mathcal{N}^T$, is enforced using constraint (11). Constraint (12) guarantees that each customer location is either assigned to be served by a truck or drone, while constraint (13) allows each drone $u \in \mathcal{U}$ to be dispatched at most once from each truck stop.

3.3. Timing decision I: departure and arrival times of truck

Constraints (14)–(18) determine the optimal departure and arrival times of truck at its stops.

$$a_{s+1}^T \geq d_s^T + R_{ij}^T - M(2 - y_{is} - y_{j,s+1}) \qquad \forall i,j \in \mathcal{F}, s \in \mathcal{S} \backslash \{s_{\hat{S}+1}\} \tag{14}$$

$$d_s^T \geq a_s^T \qquad \forall s \in \mathcal{S} \tag{15}$$

$$d_s^T \geq a_i^U + \sum_{j \in \mathcal{F}} G_j y_{js} - M(1 - \sum_{u \in \mathcal{U}} x_{iu,s-1}) \qquad \forall i \in \mathcal{N}, s \in \mathcal{S} \backslash \{s_0\} \tag{16}$$

$$d_s^T \geq d_i^U - M(1 - \sum_{u \in \mathcal{U}} x_{ius}) \qquad \forall i \in \mathcal{N}, s \in \mathcal{S} \tag{17}$$

$$d_s^T \geq e_i - M(1 - y_{is}) \qquad \forall i \in \mathcal{N}, s \in \mathcal{S} \backslash \{s_0\} \tag{18}$$

The arrival time of truck at stop $s+1 \in \mathcal{S}$ is the sum of truck departure time from stop $s \in \mathcal{S}$ and the travel time between these stops R_{ij}^T, as given by constraint (14). It should be noted that this constraint is active only if locations i and $j \in \mathcal{F}$ are selected as stops s and $s+1 \in \mathcal{S}$, respectively. Moreover, constraints (14) and (15) also facilitate subtour elimination for the truck route. The truck departure time from stop $s \in \mathcal{S}$ is the maximum of the following four events, which are captured by constraints (15)–(18), respectively: (i) arrival time of truck to stop $s \in \mathcal{S}$, (ii) recovery completion time of UAVs launched from the preceding stop, (iii) departure time of all UAVs from stop $s \in \mathcal{S}$, and (iv) service completion time at truck stop $s \in \mathcal{S}$, if it is a customer location.

3.4. Timing decision II: launching and landing times of drones

Constraints (19)–(23) determine the optimal time of launching and recovering drones at each truck stop.

$$d_i^U \geq a_s^T + \sum_{j \in \mathcal{F}} H_j y_{js} - M(1 - \sum_{u \in \mathcal{U}} x_{ius}) \qquad \forall i \in \mathcal{N}, s \in \mathcal{S} \tag{19}$$

$$d_j^U \geq a_i^U + \sum_{k \in \mathcal{F}} (G_k + H_k) y_{ks} - M(2 - x_{ius} - x_{ju,s+1}) \qquad \forall i,j \in \mathcal{N}, u \in \mathcal{U}, s \in \mathcal{S} \backslash \{s_{\hat{S}+1}\} \tag{20}$$

$$a_j^U \geq d_j^U + \sum_{u \in \mathcal{U}} R_{ijku}^U x_{jus} + V_j^U - M(3 - \sum_{u \in \mathcal{U}} x_{jus} - y_{is} - y_{k,s+1}) \qquad \forall i,k \in \mathcal{F}, j \in \mathcal{N}, s \in \mathcal{S} \backslash \{s_{\hat{S}+1}\} \tag{21}$$

$$a_i^U \geq a_{s+1}^T - M(1 - \sum_{u \in \mathcal{U}} x_{ius}) \qquad \forall i \in \mathcal{N}, s \in \mathcal{S} \backslash \{s_{\hat{S}+1}\} \tag{22}$$

$$a_j^U - d_j^U \leq \sum_{u \in \mathcal{U}} E_{ijku} x_{jus} + M(3 - \sum_{u \in \mathcal{U}} x_{jus} - y_{is} - y_{k,s+1}) \qquad \forall i,k \in \mathcal{F}, j \in \mathcal{N}, s \in \mathcal{S} \backslash \{s_{\hat{S}+1}\} \tag{23}$$

Constraint (19) ensures the dispatch time of each drone to be after the truck arrival to its stop plus the respective launch preparation time. Likewise, constraint (20) guarantees the departure of each drone $u \in \mathcal{U}$ from a truck stop to be after the sum of the following three times: drone arrival time from the preceding delivery, drone recovery time, and preparation time for the next delivery. Constraint (21) stipulates that the arrival of each drone must be after the flight duration and service time at the assigned customer location. In addition, constraint (22) ensures that the drone landing is allowed only after the truck reaches its stop. However, for each UAV sortie (from truck stop $i \in \mathcal{F}$ to deliver a package to a customer location $j \in \mathcal{N}$ then returning to rendezvous at stop $k \in \mathcal{F}$), the total flight and service duration must be within its endurance, as stated by constraint (23).

3.5. Timing decision III: customer service at truck stops

Each customer $i \in \mathcal{N}$ can be served by the truck operator or a drone. When a customer location is assigned to be a truck stop, the delivery is assumed to be performed by the truck operator. The following constraints control the beginning and end of such delivery services.

$$b_i \geq a_s^T - M(1 - y_{is}) \qquad \forall i \in \mathcal{N}, s \in \mathcal{S} \tag{24}$$

$$e_i \geq b_i + V_i^T - M(1 - \sum_{s \in \mathcal{S}} y_{is}) \qquad \forall i \in \mathcal{N} \tag{25}$$

$$b_i \leq M \sum_{s \in \mathcal{S}} y_{is} \qquad \forall i \in \mathcal{N} \tag{26}$$

$$e_i \leq M \sum_{s \in \mathcal{S}} y_{is} \qquad \forall i \in \mathcal{N} \tag{27}$$

If a customer location is assigned to be a truck stop, then the service start and end time by the truck operator are specified by constraints (24) and (25), respectively. Specifically, constraint (24) ensures that the service can begin only after the truck arrival to the respective stop $s \in \mathcal{S}$, while constraint (25) determines its end time depending on the service duration V_i^T. On the other hand, if a customer $i \in \mathcal{N}$ is not assigned to be served by a truck (i.e., $\sum_{s \in \mathcal{S}} y_{is} = 0$), then constraints (26) and (27) forces the service begin and end time to be zero.

3.6. Scheduling of tasks at truck stops

At each truck stop, it is crucial to coordinate the drone launch and recovery operation to avoid multiple drones departing/arriving at the same time. In addition, the scheduling of tasks at each stop must account for the truck operator availability. An operator can perform only one of the following tasks at a given time: (i) prepare a drone for launching, (ii) recover a drone, or (iii) deliver the respective package to the customer at truck stop. Constraints (28)–(37) control the optimization of scheduling these tasks.

$$w_{ijs} + w_{jis} \geq \sum_{u \in \mathcal{U}} x_{ius} + \sum_{u \in \mathcal{U}} x_{jus} + \sum_{u \in \mathcal{U}} x_{iu,s-1} + \sum_{u \in \mathcal{U}} x_{ju,s-1} + y_{is} + y_{js} - 1 \qquad \forall i,j \in \mathcal{N}, s \in \mathcal{S} \backslash \{s_0\}, i > j \tag{28}$$

$$w_{ijs} + w_{jis} \geq \sum_{u \in \mathcal{U}} x_{ius} + \sum_{u \in \mathcal{U}} x_{jus} - 1 \qquad \forall i,j \in \mathcal{N}, s \in \{s_0\} \subseteq \mathcal{S}, i > j \tag{29}$$

$$a_i^U \geq a_j^U + \sum_{k \in \mathcal{F}} G_k y_{ks} - M(3 - w_{jis} - \sum_{u \in \mathcal{U}} x_{iu,s-1} - \sum_{u \in \mathcal{U}} x_{ju,s-1}) \qquad \forall i,j \in \mathcal{N}, s \in \mathcal{S} \backslash \{s_0\} \tag{30}$$

$$d_i^U \geq d_j^U + \sum_{k \in \mathcal{F}} H_k y_{ks} - M(3 - w_{jis} - \sum_{u \in \mathcal{U}} x_{ius} - \sum_{u \in \mathcal{U}} x_{jus}) \qquad \forall i,j \in \mathcal{N}, s \in \mathcal{S} \tag{31}$$

$$a_j^U \geq d_i^U - M(2 + w_{jis} - \sum_{u \in \mathcal{U}} x_{ju,s-1} - \sum_{u \in \mathcal{U}} x_{ius}) \qquad \forall i,j \in \mathcal{N}, s \in \mathcal{S} \backslash \{s_0\} \tag{32}$$

$$d_i^U \geq a_j^U + \sum_{k \in \mathcal{F}} (G_k + H_k) y_{ks} - M(3 - w_{jis} - \sum_{u \in \mathcal{U}} x_{ju,s-1} - \sum_{u \in \mathcal{U}} x_{ius}) \qquad \forall i,j \in \mathcal{N}, s \in \mathcal{S} \backslash \{s_0\} \tag{33}$$

$$a_i^U \geq e_j - M(3 - w_{jis} - \sum_{u \in \mathcal{U}} x_{iu,s-1} - y_{js}) \qquad \forall i,j \in \mathcal{N}, s \in \mathcal{S} \backslash \{s_0\} \tag{34}$$

$$b_i \geq a_j^U + \sum_{k \in \mathcal{F}} G_k y_{ks} - M(3 - w_{jis} - \sum_{u \in \mathcal{U}} x_{ju,s-1} - y_{is}) \qquad \forall i,j \in \mathcal{N}, s \in \mathcal{S} \backslash \{s_0\} \tag{35}$$

$$b_i \geq d_j^U - M(3 - w_{jis} - y_{is} - \sum_{u \in \mathcal{U}} x_{jus}) \qquad \forall i,j \in \mathcal{N}, s \in \mathcal{S} \tag{36}$$

$$d_i^U \geq e_j + \sum_{k \in \mathcal{F}} H_k y_{ks} - M(3 - w_{jis} - \sum_{u \in \mathcal{U}} x_{ius} - y_{js}) \qquad \forall i,j \in \mathcal{N}, s \in \mathcal{S} \tag{37}$$

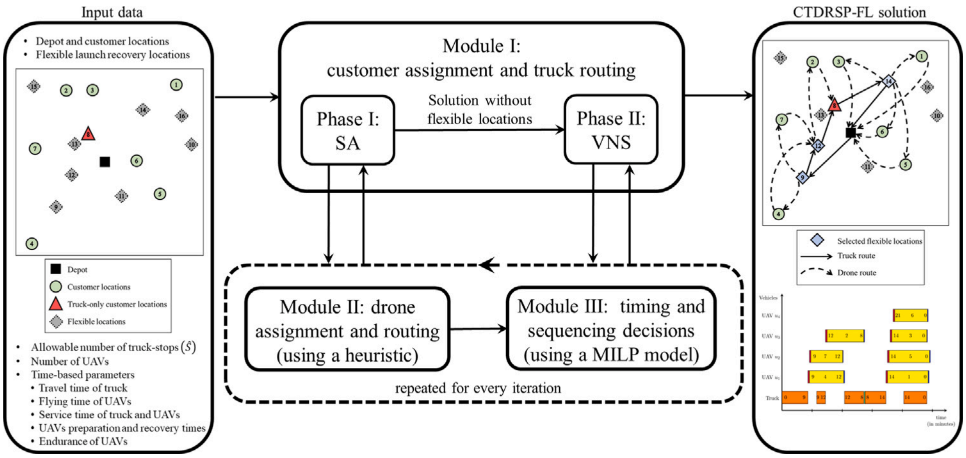

Fig. 2. Overview of the developed OTS algorithm.

The sequencing of launching and landing operations of drones as well as delivery service tasks of customers at truck stops is controlled using constraints (28) and (29) for each pair of customer locations i and $j \in \mathcal{N}$ at truck stop $s \in \mathcal{S}$. In particular, constraint (28) stipulates sequencing the pertaining tasks of locations i and $j \in \mathcal{N}$ only when they are served (by drone launching/retrieval or delivery by the truck operator) at the same truck stop, while constraint (29) similarly handles the case of launching drones from the depot. If multiple drones are recovered at a truck stop $s \in \mathcal{S}$, constraint (30) ensures collision avoidance by determining the arrival times of drones such that a drone can arrive only after the preceding drone has been recovered. Similarly, if multiple drones are launched from a truck stop $s \in \mathcal{S}$, constraint (31) specifies the launching times of drones while considering a preparation duration between each consecutive sorties. For each pair of drones, with one recovered and the other launched at the same truck stop $s \in \mathcal{S}$, constraints (32) and (33) determine their landing and launching times, respectively, while considering a recovery and preparation duration if the landing is before the launching task. When a customer location $i \in \mathcal{N}$ is assigned to a truck stop $s \in \mathcal{S}$, constraints (34)–(37) control the start and end times of delivery by the truck operator while accounting for the recovery and launching tasks of drones at the same truck stop.

4. Optimization-enabled two-phase search algorithm

The problem considered in this paper is an extension of the classical TSP, which is an NP-hard combinatorial optimization problem and cannot be solved in polynomial time. As a result, the MILP model for CTDRSP-FL becomes computationally intractable for large instances. Hence, in this section, we develop an optimization-enabled two-phase search (OTS) algorithm that decomposes the problem into three modules to handle the different decisions associated with the CTDRSP-FL. The goal of Module I is to establish the truck route, which implicitly determines the assignment of customers to be served either by a drone or truck (i.e., assignment and truck routing decisions). Module II assigns a specific drone $u \in \mathcal{U}$ to each customer who must be served by a drone (based on Module I), and establishes the drone sorties, while Module III aims to optimize the scheduling decision at each truck stop.

Fig. 2 provides an overview of the proposed algorithm and highlights the interaction between the three modules. We develop two metaheuristics, simulated annealing (SA) and variable neighborhood search (VNS), which are solved sequentially to explore and exploit the search space associated with Module I decisions. In particular, the SA determines a feasible truck route by considering only the customer locations in the delivery network, thereby restricting the search to a limited region of the CTDRSP-FL solution space. Subsequently, the VNS leverages the solution obtained by the SA and widens the search to flexible truck stop locations. Hence, the proposed two-phase sequential search approach initially focuses on a subset of the feasible solutions and then considers the entire search space to improve the solution quality. The motivation for such a two-phase sequential search approach is to obtain a good solution in a reasonable time by progressively focusing on different parts of the solution space. Furthermore, an additional benefit of our modeling approach is that the second phase can be easily integrated into existing models proposed in the literature to relax their restrictions on LARO to be at customer locations only (see Table 1).

Given the candidate solution of Module I at each iteration of the search process, Module II identifies which drone in the set $\mathcal{U}$ should be deployed to serve each customer (who were selected in Module I to be served by drones) and subsequently determines the routing of drones. Next, Module III optimizes the scheduling of truck operator tasks at each stop using a decomposed optimization model. Thus, the proposed approach collaboratively employs the three modules as the neighborhood search of the metaheuristics advances. The following subsections provide a detailed description of the three modules.

s_0	s_1	s_2	s_3	s_4	s_5	s_6	s_7	s_8	...	...	$s_{\hat{S}}$	$s_{\hat{S}+1}$
i_0	i_5	i_{12}	i_8	i_{N+7}	i_6	i_1	i_{N+3}	i_{14}	...	...	i_{N+9}	i_{N+F+1}

Fig. 3. Solution representation of truck assignment and routing in Module I.

4.1. Module I: customer assignment and truck routing

This module jointly determines the assignment of customers to each of the two vehicle types (truck or drone) and the truck route. As discussed earlier, the problem of optimizing the truck route is NP-hard. In addition, Module I also requires the assignment of vehicles to customers, which further complicates the problem under consideration. Therefore, we employ metaheuristics (SA and VNS) to efficiently explore the search space and obtain a good solution in a reasonable time. An illustrative solution representation of Module I is exhibited in Fig. 3. It depicts the customer locations assigned to the truck, flexible truck stops selected for drone LARO (e.g., nodes i_{N+3}, i_{N+7}), and sequence in which the stops are visited. The remaining customers (e.g., nodes i_2, i_3, and i_4) that are not present in Fig. 3 are assigned to be served by drones. It is important to note that the solution representation throughout the search process must be constructed by taking into account several problem characteristics to ensure feasibility. First, a drone or truck must be assigned to serve each customer $i \in \mathcal{N}^U$, while the truck is assigned by default to serve all customers in set $\mathcal{N}^T$. Let $\mathcal{N}^{U-}$ and $\mathcal{N}^{U+}$ denote the subset of customers who are selected to be served by truck and drones, respectively, where $\mathcal{N}^{U-} \cup \mathcal{N}^{U+} = \mathcal{N}^U$. Then, the truck route must cover the set of customers $\mathcal{N}^{TR-} = \mathcal{N}^T \cup \mathcal{N}^{U-}$. In addition, the truck route must start and end its route at the depot (represented by indices $\{i_0\}$ and $\{i_{N+F+1}\}$) and may also stop at non-customer (or flexible) locations belonging to set $\mathcal{F}$. If $\mathcal{N}^{TR+}$ denotes the selected flexible truck stops, then, without loss of generality, the cardinality of the set $\mathcal{N}^{TR} = \mathcal{N}^{TR-} \cup \mathcal{N}^{TR+}$ must not exceed the maximum allowable number of truck stops $\hat{S}$, as indicated in Fig. 3.

To determine the Module I decisions, we develop a two-phase search strategy using SA and VNS (see Fig. 2). SA is a well-known probabilistic search technique developed by Kirkpatrick et al. (1983) with a mechanism to escape from local optimal solutions by probabilistically accepting low quality solutions. It has been successfully used to solve various routing problems (Lin et al., 2009; Dorling et al., 2017; Kartal et al., 2017; Gonzalez-R et al., 2020; Moshref-Javadi et al., 2020; Ji et al., 2022). Also, VNS is a popular local search technique developed by Mladenović and Hansen (1997) to improve an incumbent solution by systematically visiting multiple distant neighborhoods in the solution space (Schermer et al., 2019a; Ulmer et al., 2020; Sadati and Çatay, 2021). Therefore, a two-phase approach that employs SA to obtain a good feasible solution for a defined solution space, and then leverages VNS to improve it by considering flexible LARO sites is promising for the CTDRSP-FL, especially to achieve a good trade-off between solution quality and runtime.

To generate the neighborhood solutions at each iteration of the search process, we consider three neighborhood operators for both SA and VNS. Algorithm 1 shows the procedure for sequentially generating the three neighborhood solutions. The first search operator (B_1) considers a random swap of two truck stops (lines 4–9). Given the current truck route in the search algorithm (B_0) and truck travel time ($R^T_{ij} \quad \forall i, j \in \mathcal{N}$), two random indices ($i^{\text{stop}}$ and i^{location}) are generated (line 4), one for position of the truck route (i.e., an index from set $\mathcal{S}$) and another for a potential physical location (i.e., an index from set $\mathcal{N}$) to be included in the resultant truck route. If the chosen truck stop (i.e., i^{stop}) is a customer location that must be visited by truck (i.e., $B_0[i^{\text{stop}}] \in \mathcal{N}^T$), then the swapping operation is confined to be within the existing truck route (lines 5–7), otherwise the originally chosen location (i.e., i^{location}) is considered as shown on line 9. The second operator (B_2) rearranges B_1 using the TSP nearest neighbor algorithm (Hurkens and Woeginger, 2004) (lines 11 – 13), while the third neighborhood search operator (B_3) reverses the truck route of B_2 (lines 14 and 15). Finally, the three neighborhood solutions B_1, B_2, and B_3 are returned.

4.1.1. Constructing routing solution without flexible locations

The pseudocode of the SA is shown in Algorithm 2. The SA parameters are given as inputs, namely initial and final temperature (T^0 and $T^{\min}$), cooling rate (K), number of iterations at each temperature level ($\text{iter}^{\max}$) and maximum runtime ($\tau^{\max}_{SA}$), in addition to an initial feasible truck routing solution to initiate the neighborhood search. Upon initializing the current values of the SA algorithm (lines 3–5), the loop on lines 6–20 explores the solution space over different temperature levels until one of the following two termination criteria is met: (i) current temperature level (T^{current}) falls below $T^{\min}$ or (ii) current runtime (τ) exceeds $\tau^{\max}_{SA}$. For each level, a predefined number of search iterations is conducted by loop 7–18. In each iteration, three neighbors to the current best solution are generated using Algorithm 1 (line 8). Next, the algorithms of Modules II and III (see Fig. 2) are executed to obtain fully defined solutions of the three neighbors (line 9), and the best among them is identified (line 10). If the delivery completion of the best neighborhood solution ($\hat{t}'$) is better than the incumbent solution's objective value ($\hat{t}^*$), then the incumbent solution is updated with the best neighborhood solution (along with the associated objective value), as given by lines 11 and 12. Besides, if the best neighbor outperforms the current best solution, the former replaces the latter (lines 13 and 14). Otherwise, for the purpose of enhancing the exploration of SA algorithm, the selected neighbor may be considered as the current best solution with a probability based on the metropolis acceptance criterion (lines 15–18) (Bertsimas and Tsitsiklis, 1993). Based on the truck route obtained by the SA algorithm (B^*), the customers are categorized into $\mathcal{N}^{U-}$ and $\mathcal{N}^{U+}$ (line 21). Finally, both customer assignments and truck route are returned. The obtained truck route is then given as an input to the VNS in the second phase of Module I (as shown in Fig. 2) for improvement by allowing truck stops to be at flexible locations.

Algorithm 1 Neighborhood search operators

1: **Input:** B_0 and $R^T_{ij} \quad \forall i, j \in \mathcal{N}$
2: **Output:** truck routes of three neighbors
3: $B_1 \leftarrow B_0$
4: generate two random indices, $i^{\text{stop}} \sim U(1, \hat{S})$ and $i^{\text{location}} \sim U(1, N)$
5: **if** $B_1[i^{\text{stop}}] \in \mathcal{N}^T$ **then**
6:     generate a random index, $j^{\text{stop}} \sim U(1, \hat{S})$
7:     swap $B_1[i^{\text{stop}}]$ with $B_1[j^{\text{stop}}]$
8: **else**
9:     swap $B_1[i^{\text{stop}}]$ with i^{location}
10: $B_2[s_0] \leftarrow i_0$
11: **for** $s \in \mathcal{S}$ **do**
12:     set $j^* = \text{argmin}(R^T_{B_2[s],j} | j \in B_1 \wedge j \notin B_2)$
13:     $B_2[s+1] \leftarrow j^*$
14: **for** $s \in \mathcal{S}$ **do**
15:     $B_3[s] \leftarrow B_2[\hat{S} - s]$
16: **return** B_1, B_2, B_3

Algorithm 2 Module I – phase I (SA)

Input: T^0, $T^{\min}$, K, iter$^{\max}$, and $\tau^{\max}_{SA}$
Output: customer assignments and truck route
initialize a solution B and compute its objective value $\hat{t}(B)$
set the current best solution ($B^* \leftarrow B$) and objective value ($\hat{t}^* \leftarrow \hat{t}(B)$)
$T^{\text{current}} \leftarrow T^0$
while $T^{\text{current}} \geq T^{\min}$ **and** $\tau < \tau^{\max}_{SA}$ **do**
    for iter := 1 **to** iter$^{\max}$ **do**
        generate three neighbors (B_1, B_2, B_3) to current solution B (using Algorithm 1)
        run Modules II and III (Algorithms 4 and 5) sequentially for B_1, B_2 and B_3
        set $B' = B_{\text{argmin}(B_1,B_2,B_3)}$ and $\hat{t}' = \min(\hat{t}(B_1), \hat{t}(B_2), \hat{t}(B_3))$
        if $\hat{t}' \leq \hat{t}^*$ **then**
            $B^* \leftarrow B'$ and $\hat{t}^* \leftarrow \hat{t}'$
        if $\hat{t}' \leq \hat{t}(B)$ **then**
            $B \leftarrow B'$ and $\hat{t}(B) \leftarrow \hat{t}'$
        else
            generate rand_num $\sim U(0, 1)$
            if rand_num $\leq e^{(\frac{\hat{t}' - \hat{t}(B)}{\hat{t}'} \times 100)/T^{\text{current}}}$ **then**
                $B \leftarrow B'$ and $\hat{t}(B) \leftarrow \hat{t}'$
    $T^{\text{current}} \leftarrow K \times T^{\text{current}}$
    update elapsed time (τ)
$\mathcal{N}^{U-} = \{i \in \mathcal{N}^U | i \in \mathcal{B}^*\}$ and $\mathcal{N}^{U+} = \{i \in \mathcal{N}^U | i \notin \mathcal{B}^*\}$
return $\mathcal{N}^{U-}$, $\mathcal{N}^{U+}$, B^* and $\hat{t}^*$

4.1.2. Improving routing solution with flexible locations

The VNS procedure adopted in this research is given in Algorithm 3. In addition to using Module I outputs as parameters (B^*, $\hat{t}^*$), the VNS procedure also requires the following inputs: number of iterations for exploring near and distant neighbors (iter$^{\max}_1$ and iter$^{\max}_2$) and runtime threshold ($\tau^{\max}_{VNS}$). The VNS is comprised of two search levels (loops on lines 4–11 and 6–11) that are executed for iter$^{\max}_1$ and iter$^{\max}_2$ iterations, respectively. The first level generates a single neighbor to the current best solution (line 5), while the second level widens the search by using that neighbor in generating multiple distant neighbors from the incumbent solution (lines 7–9). The random swap operator (in Algorithm 1) is used in the first level after allowing truck stops to be at flexible locations. Specifically, on line 4 of Algorithm 1, the randomly generated potential location is modified to be $i^{\text{location}} \sim U(1, N+F)$. In the second level (i.e., loop 6–11 of Algorithm 3), six neighbors are generated (line 8). Three of them are obtained by Algorithm 1, while the other three are created by adapting Algorithm 1 to only allow non-customer locations to be included (i.e., $i^{\text{location}} \sim U(N+1, N+F)$ on line 4 of the algorithm). Subsequently, the fully defined solutions of the six neighbors are obtained by running Algorithms 4 and 5 sequentially (i.e., Modules II and III as shown in Fig. 2), and the best among them is identified (lines 8 and 9). If the best neighbor outperforms the incumbent solution, the former replaces the latter (lines 10 and 11). The algorithm is terminated if the current runtime (τ) goes beyond $\tau^{\max}_{VNS}$. Finally, the best found customer assignment ($\mathcal{N}^{U-}$ and $\mathcal{N}^{U+}$) and improved truck route obtained by allowing stops at flexible locations (B^* and $\hat{t}^*$) are returned (line 13).

Algorithm 3 Module I – phase II (VNS)

1: **Input:** B^*, $\hat{t}^*$, $\text{iter}^{\max}_1$, $\text{iter}^{\max}_2$, $\tau^{\max}_{VNS}$
2: **Output:** customer assignments and truck route
3: **while** $\tau < \tau^{\max}_{VNS}$ **do**
4:     **for** iter1 $:= 1$ **to** $\text{iter}^{\max}_1$ **do**
5:         generate a neighbor (B^{l1}_1) around the current best solution (using Algorithm 1)
6:         **for** iter2 $:= 1$ **to** $\text{iter}^{\max}_2$ **do**
7:             generate neighbors ($B^{l2}_1, B^{l2}_2, \dots, B^{l2}_6$) to B^{l1}_1
8:             run Modules II and III (Algorithms 4 and 5) sequentially for $B^{l2}_1, B^{l2}_2, \dots, B^{l2}_6$
9:             set $B = B_{\text{argmin}(B^{l2}_1, B^{l2}_2, \dots, B^{l2}_6)}$ and $\hat{t}^B = \min(\hat{t}^{B^{l2}_1}, \hat{t}^{B^{l2}_2}, \dots, \hat{t}^{B^{l2}_6})$
10:             **if** $\hat{t}^B \le \hat{t}^*$ **then**
11:                 $B^* \leftarrow B$, $\hat{t}^* \leftarrow \hat{t}^B$
12: $\mathcal{N}^{U-} = \{i \in \mathcal{N}^U | i \in \mathcal{B}^*\}$ and $\mathcal{N}^{U+} = \{i \in \mathcal{N}^U | i \notin \mathcal{B}^*\}$
13: **return** $\mathcal{N}^{U-}$, $\mathcal{N}^{U+}$, B^* and $\hat{t}^*$

4.2. Module II: drone assignment and routing

For each tested truck route in Module I, Module II is used to identify the drone-customer assignment and drone sorties (see line 9 in Algorithm 2 and line 8 in Algorithm 3). Such an optimization problem can be viewed as a variant of the generalized assignment problem (a well-known NP-hard combinatorial optimization problem) between pairs of drones and customers. Owing to the problem complexity, a rule-based heuristic is developed for Module II such that the truck waiting and drone hovering are minimized. In particular, a priority is given to drone sorties that do not require truck waiting and have minimal drone hovering, and then the sorties are assigned in ascending order of truck waiting duration. A pseudocode of the developed heuristic is provided in Algorithm 4. First, a list of all feasible drone sorties is established, where each sortie is identified by a tuple (s, j, u) (line 3). Subsequently, a preprocessing step is conducted to calculate the actual truck travel time (A^T_s) and drone travel times (A^U_{sju}) from each truck stop based on the partial solution constructed in Module I (lines 4 and 5). The heuristic rule of minimizing truck waiting and drone hovering is developed in the loops of lines 6–13 and 14–21. First, the drone sorties that can be performed without truck waiting are considered (lines 6–13). If a customer can be served by only one feasible drone sortie, then that sortie is assigned by default (lines 7 and 8). Otherwise, the assignments are selected such that the time delay between the truck and drone arrival at each truck stop is minimized (lines 11 and 12). In both cases, any assigned sortie and its mutually exclusive sorties are removed from the list $\mathcal{L}$ (lines 9 and 13). Specifically, once a customer location is assigned to a drone, it should not be served again by a drone launched from a different truck stop. Furthermore, the assigned drone and launching truck stop cannot be selected again to serve a different customer. Similar to the loop on lines 6-13, assignments of drone sorties are applied until all customers are served (lines 14-21). Finally, the assignments of drone sorties and their routing are returned (line 22).

4.3. Module III: scheduling decisions

The mathematical programming model with objective function (1) and constraints (2)–(37) is decomposed in Module III to independently optimize the scheduling decisions at each truck stop. The tasks to be optimized at each truck stop $s \in \mathcal{S}$ are launching and landing of drones. Besides, if the truck stop is at a customer location, then the truck operator's service start and end time for that location must also be sequenced. Therefore, the optimization problem of Module III is analogous in some respects to the single machine scheduling problem, where the machine (equivalent to the truck operator in this paper) can process only one job (package delivery or drone launch/recovery) at a time. Moreover, the decisions of Module III involve constraints such as respecting the flight ranges of drones, which further complicate the optimization problem. To model the scheduling decisions at each truck stop as an optimization problem, the following three mutually exclusive subsets of locations are identified for each truck stop $s \in \mathcal{S}$: (i) $\mathcal{N}^+_s$ is the subset of customer locations that are visited by drones to be recovered at truck stop $s \in \mathcal{S}$, (ii) $\mathcal{N}^0_s$ is the subset containing the customer location that is assigned to be the truck stop $s \in \mathcal{S}$, otherwise $\mathcal{N}^0_s = \phi$, and (iii) $\mathcal{N}^-_s$ is the subset of customer locations to be visited by drones launched from truck stop $s \in \mathcal{S}$.

Algorithm 5 provides the pseudocode of Module III. The outputs of Modules I and II are used as inputs and the decomposed model is then solved sequentially for each truck stop $s \in \mathcal{S}$. In the decomposed model, the binary variables y_{is} and x_{ius} are fixed (lines 8 and 9) based on the truck routing and drone-customer assignments in Modules I and II, respectively, as illustrated in Fig. 2. The arrival times of both truck and drones are fixed based on the previous run of the decomposed model (lines 10 and 11). Furthermore, the sequencing binary variable w_{ijs} is limited to customer locations in sets $\mathcal{N}^+_s$, $\mathcal{N}^0_s$, and $\mathcal{N}^-_s$ and to the truck stop $s \in \mathcal{S}$ for each run of the decomposed model. Therefore, the maximum number of binary variables in each model run does not exceed $(2 \times U + 1)^2$. For each run, the arrival times of truck and drones are stored (lines 12 and 13) to be used in the subsequent model run. Finally, a fully defined solution to the delivery problem is obtained at the end of Module III. An illustrative solution representation of Module

Algorithm 4 Module II – assignment of drones to customer locations and their routing

Input: B^* and time-based parameters
Output: assignments and routing of drones
initialize a list of all feasible sorties $\mathcal{L} = (s, j, u) \quad \forall s \in \mathcal{S}\backslash\{s_{\hat{S}+1}\}, j \in \mathcal{N}^U, u \in \mathcal{U}$
determine actual truck travel time (A_s^T) from each stop $s \in \mathcal{S}\backslash\{s_{\hat{S}+1}\}$ to the next stop
calculate travel time of each done (A_{sju}^U) if dispatched $s \in \mathcal{S}\backslash\{s_{\hat{S}+1}\}$ to serve $j \in \mathcal{N}^U$ and return to the next truck stop
while a drone sortie $(s, j, u) \in \mathcal{L}$ can be performed without truck waiting **do**
  if location $j \in \mathcal{N}^U$ can be served only by a single feasible sortie in $\mathcal{L}$ **then**
    $x_{jus}^* \leftarrow 1$
    remove the corresponding sortie from the list $\mathcal{L}$
  else
    $(s^*, j^*, u^*) \leftarrow \text{argmin}_{s\in\mathcal{S}\backslash\{s_{\hat{S}+1}\}, j\in\mathcal{N}^U, u\in\mathcal{U}}(A_s^T - A_{sju}^U|_{A_s^T \geq A_{sju}^U})|_{(s,j,u)\in\mathcal{L}}$
    $x_{j^*u^*s^*}^* \leftarrow 1$
    remove sorties $(s, u, j^*) \quad \forall s \in \mathcal{S}\backslash\{s_{\hat{S}+1}\}, u \in \mathcal{U}$ and $(s^*, u^*, j) \quad \forall j \in \mathcal{N}^U$ from the list $\mathcal{L}$
while at least one customer $j \in \mathcal{N}^U$ is not served **do**
  if location $j \in \mathcal{N}^U$ can be served only by a single feasible sortie in $\mathcal{L}$ **then**
    $x_{jus}^* \leftarrow 1$
    remove the corresponding sortie from the list $\mathcal{L}$
  else
    $(s^*, j^*, u^*) \leftarrow \text{argmin}_{s\in\mathcal{S}\backslash\{s_{\hat{S}+1}\}, j\in\mathcal{N}^U, u\in\mathcal{U}}(A_{sju}^U - A_s^T|_{A_{sju}^U \geq A_s^T})|_{(s,j,u)\in\mathcal{L}}$
    $x_{j^*u^*s^*}^* \leftarrow 1$
    remove sorties $(s, u, j^*) \quad \forall s \in \mathcal{S}\backslash\{s_{\hat{S}+1}\}, u \in \mathcal{U}$ and $(s^*, u^*, j) \quad \forall j \in \mathcal{N}^U$ from the list $\mathcal{L}$
return $x_{jus}^* \quad \forall j \in \mathcal{N}, u \in \mathcal{U}, s \in \mathcal{S}$

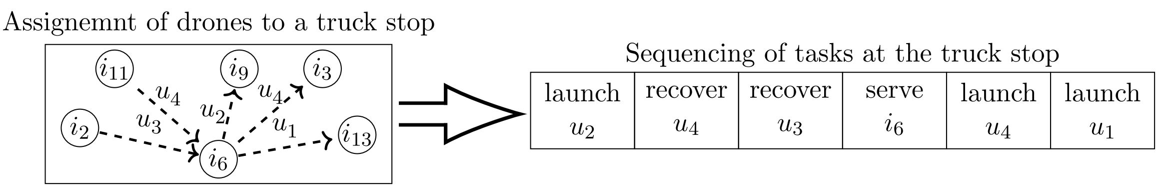

Fig. 4. An illustrative solution representation of sequencing decisions in Module III.

III at a truck stop is shown in Fig. 4, where $\mathcal{N}_s^+ = \{i_2, i_{11}\}$, $\mathcal{N}_s^0 = \{i_6\}$, and $\mathcal{N}_s^- = \{i_3, i_9, i_{13}\}$. Given the assignment of drones (both recoveries and launches) to a truck stop (i.e., customer location i_6 in the illustrative example), the tasks are sequenced as following: launch drone u_2 → recover drone u_4 → recover drone u_3 → serve customer i_6 → launch drone u_4 → launch drone u_1.

Algorithm 5 Module III – scheduling of delivery tasks

Input: B^* and $x_{jus}^* \quad \forall j \in \mathcal{N}, u \in \mathcal{U}, s \in \mathcal{S}$
Output: fully defined solution for the delivery problem
define binary parameter $y_{is}^* \quad \forall i \in \mathcal{F}, s \in \mathcal{S} \quad$ based on B^*
for each truck stop $s' \in \mathcal{S}$ **do**
  minimize objective function (1)
  $S.t.$
    constraints (2)–(34) by updating set $\mathcal{S}$ to $\mathcal{S} = \{s'\}$
    $x_{jus} = x_{jus}^* \quad \forall j \in \mathcal{N}, u \in \mathcal{U}, s \in \{s'\} \subseteq \mathcal{S}$
    $y_{is} = y_{is}^* \quad \forall i \in \mathcal{F}, s \in \{s'\} \subseteq \mathcal{S}$
    $a_s^T = a_s^{T*} \quad \forall s \in \{s'\} \subseteq \mathcal{S}$
    $a_i^U = a_i^{U*} \quad \forall i \in \mathcal{N}_{s'}^+$
  $a_{s'+1}^{T*} \leftarrow a_{s'+1}^T$
  $a_i^{U*} \leftarrow a_i^U \quad \forall i \in \mathcal{N}_{s'}^-$
return fully defined solution

5. Numerical analysis

This section includes extensive computational experiments to evaluate the impact of adopting flexible drone launch and recovery sites. The MILP model has been developed on GAMS 30.1 and solved using CPLEX 12.9. The proposed OTS algorithm is implemented in Python 3.6. All experiments were performed on a computer with an AMD Ryzen $7-2700X$ @ 3.7 GHz processor and 16 GB RAM. First, the global optimal solutions are obtained for small size problems using the proposed MILP model. We also use these instances to benchmark the performance of the OTS algorithm. Next, we demonstrate the capability of the proposed solution approach to solve relatively large-sized problems. Finally, a comprehensive analysis is conducted on large instances to assess the impact of three critical parameters, namely, distribution of flexible sites, UAV fleet size, and flight range. In order to quantify the impact of considering flexible locations, we benchmark its performance against the problem that assumes truck stops to be at customer locations only (i.e., $F = 0$), which is referred to as the collaborative truck–drone routing and scheduling problem with restricted drone launch and recovery locations (CTDRSP-RL). Note that the proposed OTS algorithm is adapted to solve large instances of CTDRSP-RL by restricting the search of VNS to customer-only locations.

5.1. Experimental setup of problem instances and OTS parameters

Test instances are created by generating randomly distributed customer and flexible stop locations on a delivery region. We consider three network configurations with 8, 25 and 50 customers distributed within areas of 15×15, 25×25, and 35×35 miles^2, respectively. Besides, we consider 10% of customers to be served only by a truck, i.e., $|\mathcal{N}^T| = \lceil 0.1 \times N \rceil$. For each problem size (N), the number of flexible locations (F) is set as $0.5N$, N, and $2N$ in our initial analysis. Furthermore, for each problem configuration (delivery locations, flexible stops, truck-only customers), we randomly generate and solve 10 instances. For each instance, the fleet of delivery vehicles is assumed to comprise a truck carrying four drones. The maximum number of allowable truck stops ($\hat{S}$) is fixed at $\lfloor N/2 \rfloor$. Similar to prior works, the velocity of both truck and drones is set at 25 mph (Murray and Chu, 2015; Ha et al., 2018; Wang and Sheu, 2019), and the drone flight range is considered to be 12 miles (Agatz et al., 2018). The travel time of vehicles and drone endurance are computed based on the coordinates of customer and flexible locations. Specifically, the truck travel time is determined by considering the rectilinear distance between nodes, while the drone travel time is calculated based on Euclidean distance. The motivation for such an approach is to emulate the truck path on a road network and the drone route in the low-altitude airspace (Ha et al., 2018). Customer service times by truck operator and drone are set at 0.5 and 1 minute, respectively (Murray and Raj, 2020). Furthermore, for all numerical experiments, we compare the performance of CTDRSP-FL with CTDRSP-RL in terms of the (i) average percentage reduction in delivery completion time (referred to as saving percent), (ii) improvement in drone utilization (η^{IMP}), which is computed as the percentage improvement in the number of customers that are served by drones instead of truck, and (iii) percent of truck stops that are selected at flexible sites (S^F).

Since the first phase of the OTS (i.e., the SA algorithm) is used to provide a good initial solution (i.e., customer assignments and truck route without flexible stops) for the second phase, its runtime is limited to 5 min. For the second phase of the OTS (i.e., the VNS algorithm), the runtime limit is set to 60 min for investigating the trade-off between solution quality and computational time (i.e., determining the average runtime required to achieve a specific solution quality for each problem size, N). In particular, the numerical results in Section 5.3 illustrate the convergence patterns for different problem sizes and aid in selecting the proper runtime threshold to be used in practice for each problem size. The other key parameters of SA and VNS algorithms are tuned by considering a representative set of instances using CALIBRA, a well-established fine-tuning procedure employing a Taguchi's fractional factorial experimental design combined with a local search (Adenso-Díaz and Laguna, 2006). The ranges explored for T^0, K, and iter^{max} (SA-related parameters) are [100, 1000], [0.8, 0.99], and [10, 30], respectively. The best performance is obtained for values: T^0 = 663, $K = 0.94$, and iter^{max} = 23. Similarly, the values explored for the number of iterations of the VNS algorithm (iter$_1^{\max}$ and iter$_2^{\max}$) is also between [10, 30], and the selected values are 13 and 15 iterations, respectively.

5.2. Results for small test instances

The results of test instances with 8 customers for both the MILP model and OTS algorithm are presented in Table 2. The proposed models are solved for the new variant (CTDRSP-FL) and the benchmark problem (CTDRSP-RL). When employing the MILP model, all the instances converged to proven optimal solutions. It is evident that the consideration of flexible sites for drone LARO can yield substantial savings in the delivery completion time as opposed to restricting them to customer locations. Furthermore, a higher reduction in delivery completion time is observed as the number of potential sites for flexible drone LARO is increased. This could be because the feasible solution space is larger when the number of potential flexible locations is varied from $0.5N$ to $2N$, thereby increasing the likelihood of establishing a flexible stop that facilitates better synchronization of truck and drones to serve the nearby customers. On average, the package delivery is 14.6% faster for the newly introduced variant when the number of flexible sites is set at $2N$. Besides, the CTDRSP-FL achieved lower delivery completion for the majority of tested instances with improvements reaching up to 27.5%. Also, the gap in Table 2 denotes the percentage deviation between the solution obtained by the OTS algorithm and global optimal. For both CTDRSP-RL and CTDRSP-FL, the proposed OTS algorithm is able to achieve the global optimal for the majority of cases, and near-optimal solution in other instances. The average optimality gap is less than 4% for both CTDRSP-RL and CTDRSP-FL. With regard to the computational performance, as shown in Table 2, the average runtime to obtain the optimal solutions using the MILP model increases exponentially when considering more flexible sites in the delivery network. For the OTS algorithm, the results in Table 2 were obtained in at most 189 s, on average, irrespective of the number of flexible sites. For instance,

Table 2
Average results of instances with $N = 8$ using the MILP model and OTS algorithm.

Problem	F	Delivery completion time			Saving%	Runtime (in s)		
		MILP	OTS	Gap%		MILP	OTS	Gap%
CTDRSP-RL	–	109.6	110.0	0.4%	–	36	130	261%
CTDRSP-FL	0.5N	103.3	107.0	3.6%	5.7%	158	132	−16%
	N	100.9	102.8	1.9%	7.9%	728	131	−82%
	2N	93.6	94.0	0.4%	14.6%	2,695	189	−93%

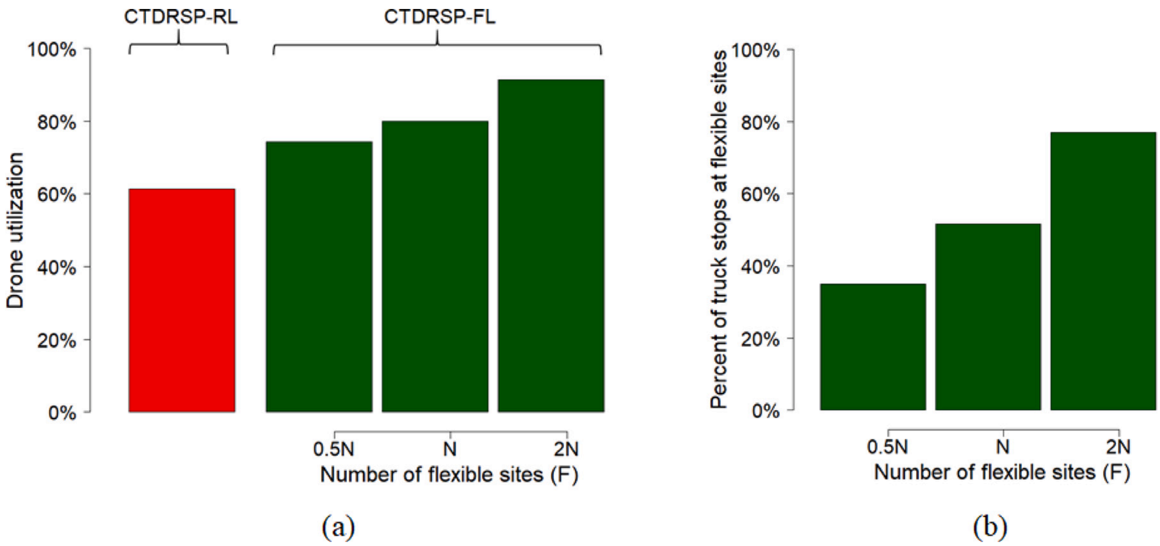

Fig. 5. Characteristics of optimal solutions for $N = 8$: (a) drone utilization of CTDRSP-RL and CTDRSP-FL; and (b) percentage of truck stops at flexible sites for CTDRSP-FL.

when F is set at $2N$, all instances in the first phase converged to the minimum temperature within 300 s (min: 62, average: 171, max: 288), while the second phase found the best solutions (which are optimal for most instances) in less than 60 s (min: 4, average: 17, max: 52).

Fig. 5 shows the characteristics of the optimal solutions for $N = 8$. In Fig. 5(a), the improvement in drone utilization when considering flexible truck stops is shown as opposed to restricting them to customer locations. The drone utilization is calculated as the ratio of customers served by drones to the total number of droneable locations. It can be observed that the percentage improvement in drone utilization increases when F is increased. Since drones have limited flight ranges, restricting their LARO to customer-only locations may limit the number of feasible drone sorties. Hence, increasing the number of flexible stops improves the likelihood of deploying drones, while shortening the truck route and reducing its waiting time. For the majority of cases under consideration, drone utilization is better for the proposed CTDRSP-FL. This may be because of the fact that an additional customer must be visited by a drone every time a flexible site is used as a truck stop instead of a customer location, thereby engaging the drones in more sorties. Moreover, a flexible stop may allow parallelization of drone delivery such that the fleet synchronization at a truck stop is efficient (i.e., minimum waiting time), which, in turn, enables faster delivery completion. Fig. 5(b) illustrates that the likelihood of assigning truck stops to flexible sites increases as F is increased from $0.5N$ to $2N$. On average, for $F = 2N$, 77% of truck stops are at flexible locations. Another interesting observation in the optimal solution of CTDRSP-FL is that the flexible stops that were neither close nor far away from the depot were mostly chosen, while stops in the periphery were rarely selected. Specifically, we observed that the majority of the chosen flexible stops are within the area of 25% to 75% of the distance from the depot to the delivery area borders. This suggests that it is beneficial to shorten the truck path by employing drones to faraway customer locations.

5.3. Performance of OTS algorithm on larger instances

The proposed OTS algorithm is employed to obtain solutions for larger problem instances ($N = 25$ and 50) of both CTDRSP-RL and CTDRSP-FL, and the corresponding results are presented in Table 3. It shows that additional savings in delivery time can be achieved as more flexible sites (F) are considered. For $F = 2N$, the average savings in delivery completion time for the two problem sizes is 12.6% and 11.3%, respectively. However, in contrast to the results of $N = 8$, the average savings may decrease marginally as the problem size increases. In particular, for $F = 2N$, the average improvement achieved by CTDRSP-FL decreases by 2.0% (from 14.6% to 12.6%) when $N = 25$ instead of $N = 8$ and by only 1.3% (from 12.6% to 11.3%) when additional 25 customer locations

Table 3
Average results of instances with $N = 25$ and $N = 50$ using the OTS algorithm.

N	Problem	F	$\hat{t}$	Saving %	η^{IMP}	S^F	Runtime (in seconds)
25	CTDRSP-RL	–	296.2	–	–	–	501
	CTDRSP-FL	0.5 N	277.2	6.4%	10.7%	16.7%	942
		N	268.7	9.3%	21.4%	32.2%	1,240
		2N	259.0	12.6%	30.4%	44.0%	1,589
50	CTDRSP-RL	–	538.9	–	–	–	2,622
	CTDRSP-FL	0.5 N	503.3	6.6%	7.8%	12.0%	2,982
		N	498.1	7.6%	15.3%	22.5%	2,960
		2N	477.8	11.3%	26.9%	35.0%	2,844

Table 4
Levels of key factors affecting the impact of using flexible locations.

Factor	Levels		
	1	2	3
Distribution of customer locations	C	RC	R
Number of UAVs	2	4	6
Flight range (in miles)	12	15	20

are considered. Nevertheless, the standard deviation of the improvement achieved tends to decrease as the problem size increases from 8 to 50 customers. In other words, the percentage savings in delivery time is consistent and close to average as problem size increases, especially for 50 customers. Besides, consistent with our initial findings, the drone utilization is higher if flexible drone LARO are allowed. Therefore, even for large size problems, using flexible locations for drone LARO can achieve substantial savings in delivery completion time and better drone utilization.

With regards to the characteristics of the best solutions obtained by the OTS algorithm, we observed that when a flexible location is closer to a customer (e.g., within one mile), it is less likely to be selected as a truck stop. Moreover, if it is selected, the resulting reduction in delivery time is usually marginal. Also, we observed that a shorter truck route can yield substantial savings in delivery completion time. Therefore, flexible locations around the depot are more likely to be selected. In particular, flexible locations should neither be very near to the depot nor the periphery of the delivery area. Furthermore, analysis of the different instances shows that, in certain cases, a solution with fewer flexible locations serving as truck stops can achieve a better reduction in delivery time than a plan with more such locations. Thus, both the number of flexible truck stops as well as their relative position to the depot and customer locations are crucial for efficiently reducing delivery time in the CTDRSP-FL.

Although the runtimes in Table 3 are less than an hour, route planners may want to obtain the solutions in lesser time. Therefore, the convergence patterns of solutions with $F = 2N$, as an example, are analyzed in Fig. 6(a) to understand the improvement rates over the algorithm runtime. It can be observed that the improvement rate is fast at the beginning and decreases over time. Also, the convergence patterns for the different problem sizes are similar, but the convergence speed becomes slower with the increase in the problem size. For example, while the best solutions for $N = 8$ (which are optimal for most instances according to Table 2) could be achieved in less than 60 s, runtimes of 1600 and 2500 s were required to obtain an average gap of 0.5% to the best-known solutions for the instances with $N = 25$ and 50, respectively. Fig. 6 (b) illustrates the runtime required to achieve 3% gap to the best known solutions for the tested instances, which averages 18, 371, and 1170 s for $N = 8, 25$, and 50, respectively. Accordingly, delivery operators with limited computational resources may adopt such runtime thresholds in practice and still achieve good coordinated route plans. Thus, Fig. 6 can provide operators with insights on the trade-off between solution quality and computational time.

5.4. Impact of critical parameters on delivery completion time

The delivery completion time may be affected by the problem parameters. In this section, we assess the sensitivity of the CTDRSP-FL to three critical parameters by considering three levels for each of them, as shown in Table 4. First, the impact of different spatial distributions of customer locations is evaluated by considering three alternatives observed in the literature — (i) Clustered (C): customers are partitioned into multiple dense isolated clusters within the delivery region, (ii) Random-Clustered (RC): half of the delivery locations are randomly distributed and the remaining customers are grouped into multiple isolated clusters, and (iii) Random (R): all delivery locations are randomly positioned in the service region (Uchoa et al., 2017; Nguyen et al., 2022). Similar to prior studies, the customers are clustered by randomly locating $c \subset \mathcal{N}$ customers within the delivery region to serve as cluster seeds. The selected seeds will subsequently attract the remaining $|\mathcal{N}| - c$ customers with an exponential decay (Uchoa et al., 2017). Second, the impact of UAV fleet size per truck is assessed. Finally, longer flight ranges than those used in the baseline experiments are considered. We vary the factor levels one at a time to assess its impact, while fixing the other parameters to the baseline setting. Moreover, similar to the previous analysis, we run 10 replications for each problem configuration. Thus, a total of 180 test instances (3 factors × 3 levels × 10 replications × 2 variants) are generated and analyzed.

Table 5 provides the impact of different spatial distributions of customer locations on the delivery completion time, drone utilization, and assignment of truck stops to flexible sites. The results indicate that the proposed CTDRSP-FL achieves faster deliveries

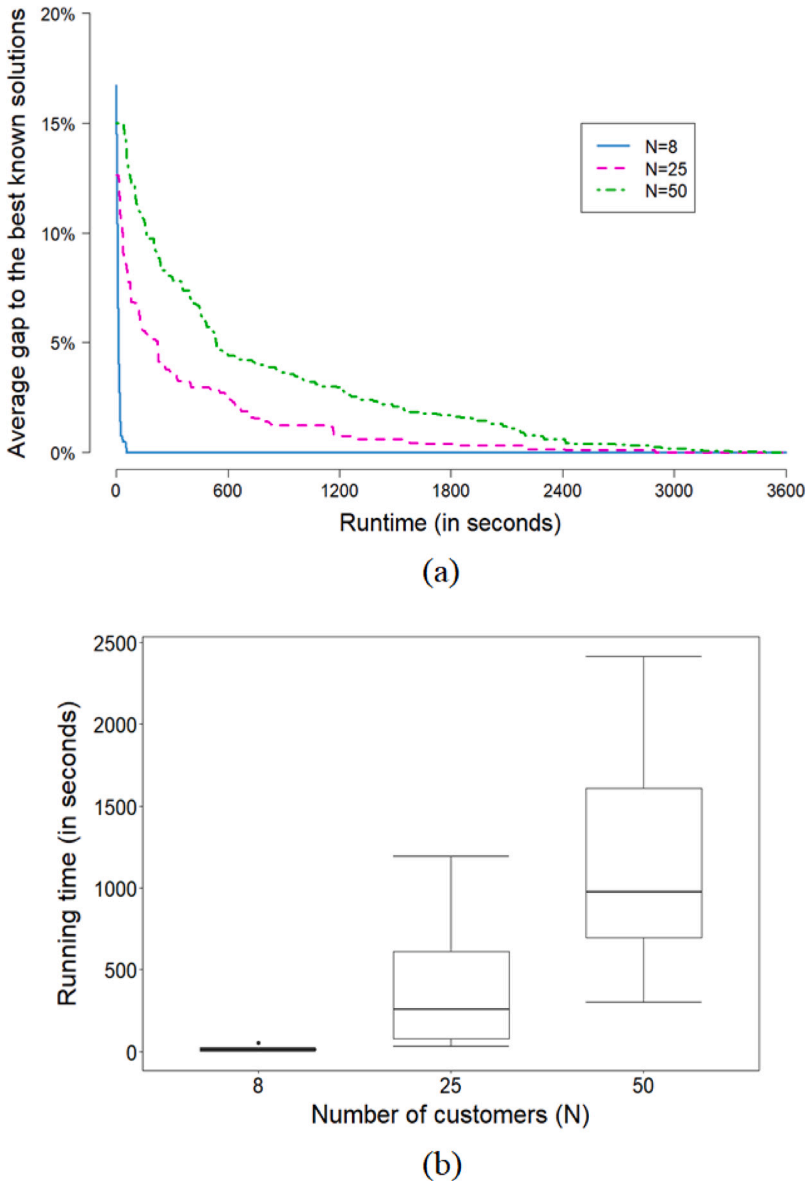

Fig. 6. Runtime results of the second phase in the OTS algorithm for different problem sizes with $F = 2N$: (a) Convergence patterns; (b) Running times to achieve 3% gap to the best known solutions.

and better drone utilization when compared to CTDRSP-RL for all the three spatial distributions under consideration. Specifically, the time saving is the highest when the customers are randomly distributed, followed by RC and C spatial distributions. When customers are grouped into multiple clusters, then the likelihood of a customer location serving as a drone launch/recovery point for the nearby locations within that cluster is high given their proximity. This could be the reason for the lower time savings in the case of clustered customer locations. Nevertheless, even modest improvements on daily route plans could have a substantial improvement over a long time period. Therefore, it is beneficial to employ flexible LARO sites irrespective of the spatial distribution of the customers.

Table 6 shows the impact of varying the drone fleet size. It is clear that operating more drones per truck can reduce the delivery completion time. The makespan obtained for CTDRSP-FL is better than CTDRSP-RL even if only two drones are employed. However, it is also observed that there is no monotonic relation between the number of drones and the corresponding savings in makespan by using flexible locations, especially for test instances with $N = 25$. Also, an important finding is that using flexible truck stops can achieve faster deliveries with fewer drones as opposed to restricting LARO to customer locations (CTDRSP-RL) and employing more

Table 5
Average results for using different levels of distribution of customer locations.

N	Distribution	Delivery completion time			η^{IMP}	S^F
		CTDRSP-RL	CTDRSP-FL	Saving %		
25	C	201.3	192.3	4.5%	14.6%	21.1%
	RC	242.0	223.1	7.8%	26.2%	37.8%
	R	296.2	259.0	12.6%	30.4%	44.0%
50	C	307.2	296.1	3.6%	18.8%	23.5%
	RC	489.6	452.4	7.6%	26.4%	33.0%
	R	538.9	477.8	11.3%	26.9%	35.0%

Table 6
Average results for using different levels of number of drones.

N	U	Delivery completion time			η^{IMP}	S^F
		CTDRSP-RL	CTDRSP-FL	Saving %		
25	2	300.1	277.6	7.5%	23.9%	34.6%
	4	296.2	259.0	12.6%	30.4%	44.0%
	6	287.9	257.9	10.4%	31.4%	48.3%
50	2	543.2	526.6	3.1%	8.4%	9.5%
	4	538.9	477.8	11.3%	26.9%	35.0%
	6	516.7	451.8	12.6%	24.4%	33.0%

Table 7
Average results for using different levels of flight range.

N	Flight range	Delivery completion time			η^{IMP}	S^F
		CTDRSP-RL	CTDRSP-FL	Saving %		
25	12	296.2	259.0	12.6%	30.4%	44.0%
	15	261.4	240.8	7.9%	31.5%	46.7%
	20	250.1	236.5	5.4%	32.3%	47.8%
50	12	538.9	477.8	11.3%	26.9%	35.0%
	15	502.3	461.8	8.1%	29.2%	37.0%
	20	486.0	454.7	6.4%	24.0%	30.5%

drones. For example, in the instances of $N = 25$, the average makespan for CTDRSP-FL with two drones is lower than CTDRSP-RL with six drones, as shown in Table 6. Likewise, for instances with 50 customers, the delivery completion time with four drones and flexible LARO is earlier than adopting six UAVs but restricting truck stops to customer locations. Furthermore, the average improvement in drone utilization with $U = 4$ and 6 is similar for both problem sizes. Thus, using flexible locations for drone LARO can aid the operators not only in reducing delivery time but also in better utilizing their available fleet of drones. It is observed that the proportion of stops that were flexible tends to be higher for $N = 25$ when compared to instances with 50 customers, as shown in Table 6. As the number of customer locations increases within a delivery region, the likelihood of choosing some of them for drone LARO also increases, which, in turn, could lead to the selection of fewer flexible stops. Also, S^F increases with the drone fleet size for instances with 25 customers, but such a trend was not observed for $N = 50$.

The reduction in delivery completion time of the two variants, η^{IMP}, and S^F achieved by employing drones with different flying ranges are presented in Table 7. Utilizing longer range drones results in faster delivery completion for both CTDRSP-RL and CTDRSP-FL. A drone with a longer range could be capable of reaching more customer locations (due to larger coverage) from a given truck stop. In other words, sorties that were not feasible with a short-range drone might now be possible. Thus, such a capability allows greater parallelization of delivery tasks and could be a reason for the observed pattern in Table 7. Nevertheless, allowing flexible truck stops always yielded superior makespan as opposed to restricting them to customer-only locations. Also, the utilization of drones is not drastically affected as the range increases. This may be because the range primarily impacts the launch and recovery locations, but not the number of customers served by the drone. Most importantly, allowing flexible stops can enable faster delivery with short-range UAVs when compared to CTDRSP-RL employing long-range drones. For example, in Table 7, the average delivery time for $N = 50$ and flight range of 12 miles is 477.8 min if flexible locations are used, which is lower than operating drones with longer flight ranges but without allowing flexible locations (i.e., CTDRSP-RL). Also, the S^F is not substantially affected by the range of drones.

6. Conclusion and future research

In this section, concluding remarks and a discussion of key findings compared to previous research are provided. Furthermore, limitations of our approach are presented with suggestions for future research directions.

6.1. Concluding remarks and key research findings

This paper addresses the problem of last-mile delivery using a fleet of heterogeneous drones working in tandem with a single truck. In contrast to most of the previous works, this paper does not restrict drone launch and recovery operations to be at customer locations. Specifically, we propose a new variant of the truck–drone tandem called the collaborative truck–drone routing and scheduling problem with flexible launch and recovery locations (CTDRSP-FL). While prior research used drones to aid the trucks in package delivery, the proposed variant seeks to better exploit the drones for deliveries while embracing the assistance from truck stopping at flexible sites (feasible non-customer locations). Furthermore, the addressed problem variant considers the scheduling of truck operator tasks at a given stop, which possibly include customer service and launch/recovery of multiple drones.

A new MILP model is formulated to optimally solve the CTDRSP-FL with the objective of minimizing delivery completion time. The proposed optimization model considers several practical aspects, such as the necessity of serving some customers by trucks and scheduling of tasks at truck stops (e.g., launching and recovery of drones and customer service by the truck operator). Furthermore, an optimization-enabled two-phase search (OTS) algorithm is developed to efficiently solve large instances. While the first phase shares many characteristics with existing studies (e.g., truck–drone routing through a network of customer nodes), the second phase is developed to serve as an improvement module that leverages the presence of flexible drone launch and recovery sites. Numerical analysis shows that substantial savings in delivery completion time (>25% in some instances) can be achieved by using non-customer locations for drone launch and recovery operations. Besides, the utilization of drones was also substantially higher for the proposed variant as opposed to the traditional truck–drone tandem. Our findings also led to several practical insights on selecting the potential flexible stops such as avoiding locations with high demand density and selecting areas that are likely to shorten the truck route.

6.2. Limitations and future research directions

Although this study has many merits, there are some limitations that could be addressed in future research. First, the proposed model and solution approach is restricted to a drone with a payload limit of one parcel per sortie. While this assumption is consistent with the actual capabilities of many commercial drones, recent efforts have led to the development of drones that can carry multiple packages (Wingcopter, 2022). Furthermore, some research works have also incorporated such characteristics for vehicle routing, and therefore future research can consider adapting the proposed approach to these existing routing models (Chen et al., 2021; Choi and Schonfeld, 2017; Poikonen and Golden, 2020; Luo et al., 2021). In addition, the addressed problem can be extended to include other characteristics such as multiple trucks and variable drone speeds. Second, the optimization of scheduling decisions in the proposed solution approach (Module III) may become computationally expensive as the number of drones per truck increases. Therefore, a polynomial time algorithm that provides good (not necessarily optimal) solutions to the scheduling problem arising at each truck stop can be considered in future works. Also, the performance of the proposed OTS algorithm can be benchmarked against other heuristic/metaheuristic approaches. Third, this research focuses on truck–drone routing by assuming the set of infrastructure locations (i.e., flexible sites) to be known. Future research could complement this work by developing models that determine the flexible stop locations by taking into account the geographical region, demand density, and depot location, and then evaluating the impact of truck–drone routing by considering these established sites. Moreover, the insights obtained from this study can also be integrated into the location selection algorithms for efficiently guiding the search process.

CRediT authorship contribution statement

Mohamed R. Salama: Conceptualization, Methodology, Data curation, Formal analysis, Investigation, Visualization, Writing – original draft, Writing – review & editing. **Sharan Srinivas:** Conceptualization, Methodology, Investigation, Supervision, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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